

LANZHOU ZHONGCHUAN, China

WMO: 525331

Lat: 36.515N

Lon: 103.620E

Elev: 6388

StdP: 11.61

Time Zone: 8.00 (E08)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-2.2	1.5	-11.6	3.7	7.9	-9.4	4.2	10.7	16.6	25.5	14.6	23.0	4.4	350	0.538

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.1	86.1	60.4	83.9	60.0	80.7	58.9	65.5	77.8	64.0	75.6	62.6	73.5	6.8	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.5	107.5	69.1	60.7	100.9	68.4	59.0	94.7	67.2	34.3	78.3	33.1	76.7	31.8	72.6	69.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
18.4	15.2	12.5	DB	-7.1	91.0	5.2	2.7	-10.9	93.0	-13.9	94.6	-16.8	96.1	-20.6	98.1	(4)
			WB	-7.9	67.6	4.9	1.5	-11.5	68.7	-14.4	69.6	-17.1	70.5	-20.7	71.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	45.3	17.6	25.1	37.7	48.8	56.9	64.4	68.2	66.5	57.2	46.5	33.1	20.5	(6)
		DBStd	18.39	6.25	7.65	7.45	6.81	5.86	4.53	4.71	5.03	4.64	5.75	6.48	5.45	(7)
		HDD50	3761	1005	697	386	99	12	0	0	4	137	507	914		(8)
		HDD65	7450	1470	1117	848	486	254	63	22	43	237	574	957	1379	(9)
		CDD50	2047	0	0	3	63	227	431	564	511	220	28	0	0	(10)
		CDD65	261	0	0	0	0	4	44	121	89	3	0	0	0	(11)
		CDH74	2974	0	0	1	33	170	604	1274	838	51	3	0	0	(12)
		CDH80	736	0	0	0	3	16	112	396	208	1	0	0	0	(13)
(14)	Wind	WSAvg	5.5	4.8	5.6	6.5	6.9	6.5	5.7	5.3	5.3	4.8	4.9	4.7	4.6	(14)
(15)	Precipitation	PrecAvg	11.5	0.1	0.1	0.3	0.6	1.4	1.7	2.4	2.4	1.7	0.7	0.1	0.0	(15)
(16)		PrecMax	14.6	0.3	0.4	0.7	2.5	3.2	3.3	4.1	3.7	2.8	1.8	0.3	0.1	(16)
(17)		PrecMin	8.7	0.0	0.0	0.0	0.0	0.3	0.7	0.4	1.0	0.6	0.1	0.0	0.0	(17)
(18)		PrecStd	1.7	0.1	0.1	0.2	0.5	0.7	0.7	1.1	0.7	0.6	0.4	0.1	0.0	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	DB	45.0	55.5	69.5	78.6	82.2	86.1	91.5	88.1	78.8	71.2	57.6	44.9		(19)
(20)		MCWB	30.4	36.3	45.0	50.1	53.6	58.6	62.7	62.4	58.3	51.1	40.7	31.3		(20)
(21)		DB	39.6	48.4	62.4	72.0	78.7	83.8	87.5	84.6	75.1	66.4	53.7	40.7		(21)
(22)		MCWB	27.6	33.7	41.2	46.8	52.2	57.0	61.7	61.5	55.7	49.7	38.6	29.3		(22)
(23)		DB	35.5	44.6	57.4	68.2	75.3	80.5	84.4	82.3	71.7	62.5	50.1	37.0		(23)
(24)		MCWB	25.3	31.6	39.1	45.7	51.2	56.2	61.0	60.9	55.2	48.0	37.4	27.3		(24)
(25)		DB	31.6	39.6	53.5	64.5	71.7	77.3	81.0	78.8	68.4	58.8	46.3	33.6		(25)
(26)		MCWB	23.3	29.2	37.1	44.2	49.9	55.3	60.6	59.9	54.8	46.4	35.4	25.3		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	WB	30.5	38.1	46.1	52.3	57.9	63.8	67.7	67.5	61.7	55.1	42.5	32.1		(27)
(28)		MCDB	44.4	52.8	63.6	68.2	72.7	73.7	81.4	79.4	71.4	64.7	53.6	42.4		(28)
(29)		WB	27.7	34.2	42.1	49.6	55.5	61.9	66.0	65.4	59.7	52.1	40.2	29.6		(29)
(30)		MCDB	39.2	47.1	58.6	66.7	69.8	72.7	79.2	76.8	68.3	61.4	50.8	39.1		(30)
(31)		WB	25.2	31.8	39.9	47.8	53.7	60.1	64.4	64.2	57.9	49.9	38.2	27.5		(31)
(32)		MCDB	34.4	43.4	55.5	64.4	68.0	71.4	77.7	75.0	66.4	58.8	48.1	36.2		(32)
(33)		WB	23.2	29.6	37.7	45.6	52.1	58.3	62.8	62.8	56.4	48.0	36.0	25.4		(33)
(34)		MCDB	30.8	39.3	52.0	60.4	66.8	70.8	75.0	72.9	65.3	56.4	44.9	32.8		(34)
(35)	Mean Daily Temperature Range	MDBR	25.5	25.0	24.8	25.1	23.8	22.9	22.1	20.6	19.6	21.2	23.0	25.3		(35)
(36)		5% DB	32.6	32.7	33.0	33.3	30.9	29.7	29.4	26.8	26.9	27.8	28.2	30.2		(36)
(37)		MCWBR	22.6	20.3	17.1	15.2	12.0	9.8	9.4	9.0	11.2	13.8	15.6	20.3		(37)
(38)		5% WB	30.7	30.7	30.0	28.3	25.3	20.7	22.4	19.7	19.8	23.3	26.3	28.5		(38)
(39)		MCWBR	21.7	19.5	15.8	13.5	10.8	8.1	8.4	8.0	9.7	12.4	15.1	19.4		(39)
(40)	Clear-Sky Solar Irradiance	taub	0.328	0.385	0.447	0.486	0.487	0.477	0.461	0.474	0.449	0.411	0.345	0.315		(40)
(41)		taud	2.149	1.989	1.855	1.805	1.875	1.963	2.084	2.022	2.036	2.048	2.154	2.209		(41)
(42)		Ebn at Noon	267	262	256	254	256	258	261	255	254	250	258	264		(42)
(43)		Edh at Noon	36	48	60	67	63	58	51	53	50	45	36	32		(43)
(44)	All-Sky Solar Radiation	RadAvg	936	1199	1528	1786	1924	1906	1846	1694	1330	1203	1006	891		(44)
(45)		RadStd	86	75	88	105	110	103	157	108	99	105	65	37		(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page